

Jessica Reiff

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Education

Iowa State University, Bachelor of Science in Electrical Engineering – Ames, IA

Expected Dec 2026

- GPA: 3.87

Professional Experience

Embedded Intern, Grace Technologies – Davenport, IA

May 2026 – Aug 2026

- Developed over-the-air (OTA) firmware update functionality for a Gecko MCU, transmitting firmware updates over UART using HTTP.
- Engineered an updates section within Forge, Grace Technologies' internal application, while working within an existing codebase and database.
- Debugged embedded communication and firmware behavior using logic analyzers and Docklight, analyzing UART traffic and identifying communication issues.
- Implemented and tested device communication using the DNP3 protocol, gaining experience with industrial networking and communications.

Embedded Intern, AthenaGTX – Johnston, IA

May 2025 – Aug 2025

- Built embedded C firmware for motion-tracking prototypes using a BMI323 IMU and BMD-350 BLE modules.
- Created an Android Java BLE test application to debug and validate embedded communication.
- Collaborated with hardware and mechanical teams on PCB development, system integration, and documentation.

MATLAB Student Ambassador, MathWorks – Ames, IA

Jan 2025 – Present

- Coordinated workshops and seminars to demonstrate MATLAB's capabilities, enhancing students' technical skills.
- Provide peer technical support and serve as a liaison between MathWorks and Iowa State University.

Undergraduate Research Assistant, Iowa State College of Engineering – Ames, IA

Mar 2025 – Present

- Program autonomous driving algorithms for the F1TENTH platform using ROS2 in simulation and real-world autonomous racecar environments.
- Design and test path planning, obstacle avoidance, and control algorithms for autonomous navigation.
- Implement sensor fusion pipelines using LiDAR, IMU, and camera data for localization and perception in autonomous systems.

Academic Project

OSCAR – Open-Source Copter for Academic Research | Firmware & Controls

Jan 2026 – Dec 2026

- Implemented embedded firmware for STM32-based quadcopter flight controller hardware using SPI, I2C, UART, and USB communication protocols.
- Applied quaternion-based orientation estimation and sensor fusion using IMU and magnetometer sensor data.
- Integrated ExpressLRS receiver communication for low-latency pilot control input.
- Configured and debugged real-time sensor pipelines and high-frequency flight control subsystems.
- Performed register-level peripheral configuration and hardware debugging using oscilloscopes and logic analyzers.

Leadership

Electrical Engineer Curriculum Chair, IEEE

Aug 2024 – Present

- Attend ECPE Curriculum meetings as representative for Electrical Engineering students.
- Coordinate with faculty and peers to provide student feedback on curriculum updates.

Technical Skills

- **Programming Languages:** C, Embedded C, C++, Python, Java, MATLAB, VHDL, Verilog, C#
- **Embedded Robotics:** FPGA, STM32, FreeRTOS, ROS2, Sensor Fusion, Quaternion Mathematics, Madgwick AHRS, Real-Time Systems, Embedded Firmware Development
- **Communication Protocols:** SPI, I2C, UART, BLE, USB, DNP3, ExpressLRS
- **Tools & Software:** STM32CubeIDE, MATLAB, Simulink, Android Studio, KiCad, QuestaSim, Git, Vivado
- **Hardware & Lab:** Oscilloscopes, Logic Analyzers, PCB Prototyping, Soldering, Circuit Debugging